

The Tonnetz spectrum is the generating triad's Fourier balance profile

a machine-checked DFT account of pitch-class symmetry,
from the chromatic circle to the neo-Riemannian Tonnetz

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Abstract

The neo-Riemannian **Tonnetz** — the graph on the 24 major and minor triads joined by the parsimonious P, L, R moves — has an adjacency spectrum, and our main result identifies it exactly: **the eigenvalues of the Tonnetz are $\pm$ the Fourier balances of the single triad that generates it**, the signed set of distinct eigenvalues $\pm\{3, \sqrt{5}, \sqrt{3}, 1, 2 \cos \frac{\pi}{12}, 2 \cos \frac{5\pi}{12}\} = \{\pm|\hat{\mathbf{1}}_T(k)|\}_{k=0}^6$ ($|a_2| = |a_6| = 1$ coincide). The augmented balance $\sqrt{5} = |a_3|$, the diminished $\sqrt{3} = |a_4|$, the whole-tone $1 = |a_6|$ — the chord's own periodicity weights — are literally the graph's vibration frequencies (Theorem 1). We prove it by building the one tool that makes it inevitable: the discrete Fourier transform is the shared eigenbasis of pitch-class symmetry. (i) every translation-invariant operator on $\mathbb{Z}_N$ (the chromatic cycle C_{12} , every abelian Cayley graph) is **diagonalized** by the DFT characters, eigenvalue the Fourier coefficient of its connection set (Babai). (ii) adjoining inversion makes the T/I group D_{2N} **block-diagonal**, pairing $a_k \leftrightarrow a_{-k}$ into conjugate-pair irreducibles. (iii) the Tonnetz is the Cayley graph of the non-abelian $PLR \cong D_{12}$ group, and reading it through (i)–(ii) yields the balance-profile identity, *complete with multiplicities*. Every step is machine-checked in Lean 4, sorry-free and axiom-clean — including the unconditional **completeness** of the spectrum (an explicit basis of eigenvectors; no Sage, no irrep classification). The spectral formula itself is standard — the dihedral-Cayley computation of Gao–Luo [10], of which the Tonnetz is the $S = \{P, L, R\}$ instance, and the Tonnetz Laplacian was studied by Lostonlen. What we add is the *reading* of that adjacency spectrum as the generating triad's balance profile, the identification of its spectral fibres with the music-theoretic Z -relation (homometric chords give cospectral graphs — Babai's old criterion, read musically), and the Lean certification. Not new mathematics — a unified, certified account.

1 The main result, and the one idea behind it

A chord is a set of pitch classes $A \subseteq \mathbb{Z}_{12}$. Encode it by its 0/1 indicator and take the discrete Fourier transform $\hat{A}$; its coefficients $a_k = \hat{A}(k)$ are the **Fourier balances** of Lewin and Quinn [18] — a_1 measures chromatic clustering, a_3 augmented content, a_4 diminished-seventh content, a_5 diatonic

*Prepared with the assistance of artificial intelligence tools. The mathematics is classical; every claim, scope statement, and limitation is the author's responsibility, and the Lean 4 kernel—not the assistant—is what certifies the formalized parts. The boundary of the contribution is stated plainly in Section 8.

content, a_6 whole-tone content (the standard dictionary, fixed by which scale maximizes each $|a_k|$). But the chords themselves also form a *graph*: the neo-Riemannian **Tonnetz**, whose 24 vertices are the major and minor triads and whose edges are the three parsimonious moves P (parallel), L (leading-tone), R (relative), each swapping one voice. This note is about a single number attached to that graph — its adjacency spectrum — and the surprise is that it is not new data at all.

Theorem 1 (Tonnetz spectrum = generating triad’s balance profile). *Let A be the adjacency operator of the neo-Riemannian PLR Tonnetz on the 24 major/minor triads, with generating triad $T = \{0, 3, 7\}$ (set class 3-11). The eigenvalues of A are exactly $\pm|\hat{\mathbf{1}}_T(k)|$ for $k = 0, \dots, 6$ — the signed set of distinct values of T ’s own Fourier balances $\pm\{3, \sqrt{5}, \sqrt{3}, 1, 2\cos\frac{\pi}{12}, 2\cos\frac{5\pi}{12}\}$ — with multiplicities recorded by the characteristic polynomial $(x-3)(x+3)(x-1)^3(x+1)^3(x^2-5)^2(x^2-3)^2(x^4-4x^2+1)^2$. The eigenvalues are machine-checked in `TonnetzSpectrum.lean`; that they are all of them (completeness) in `TonnetzCompleteness.lean`.*

Why should a chord’s harmonic colour — its augmented, diminished, whole-tone content — be the very same data as the vibration frequencies of the graph that chord generates?

Because both are read by *one eigenbasis*. The discrete Fourier characters simultaneously diagonalize every translation-invariant operator; adjoining inversion folds them into conjugate-pair blocks; and the Tonnetz, being the Cayley graph of the dihedral $PLR \cong D_{12}$ group, is keyed to exactly those blocks — so its eigenvalues can be nothing other than the generating triad’s balances. The note is that one idea, built rung by rung (§§2–5) up to Theorem 1, then its geometry (§6), its Lean certification (§7), and its honest boundary (§8). Where Note #1 [20] followed one *chord* into group theory and Note #2 [21] one *question* into the phase of $\hat{A}$, this note follows one *spectrum* back to its chord.

2 Setup: characters, circulants, and the DFT eigenbasis

Fix a universe $\mathbb{Z}_N$. Mathlib’s `ZMod.dft` (Loeffler [13]) is built on the standard additive character $\text{stdAddChar } j = e^{2\pi i j/N}$. The k -th **DFT character** is the vector $\text{dftChar } k(j) = \text{stdAddChar}(jk) = e^{2\pi i jk/N}$. These N vectors form a **basis** of $\mathbb{C}^{\mathbb{Z}_N}$ (the characters of the finite abelian group $\mathbb{Z}_N$). A **circulant** matrix v has entry $(i, j) \mapsto v(i-j)$ — a shift-invariant (convolution) operator with first column v ; every translation-invariant operator on $\mathbb{Z}_N$ is a circulant.

3 The abelian floor: circulants and Cayley graphs are diagonalized by the DFT

The keystone. Every circulant is diagonalized by the DFT characters, the eigenvalue being exactly the Fourier coefficient of its first column:

$$(\text{circulant } v) \text{dftChar } k = \hat{v}(k) \cdot \text{dftChar } k \quad (\text{circulant_mulVec_dftChar}).$$

The proof is one reindexing plus the character law; no analysis. Because the characters are a basis, the DFT is the *simultaneous* diagonalizing eigenbasis of all of harmony’s shift-invariant operators.

The chromatic cycle. The adjacency operator of C_{12} is the circulant of the ± 1 semitone indicator, so its eigenvalues collapse to the textbook $\text{cycAdj}(k) = 2\cos(2\pi k/12)$ (`C12_eigenvalue_cos`); and this C_{12} is Mathlib’s graph, `C12_eq_adjMatrix`: $C_{12} = (\text{cycleGraph } 12).\text{adjMatrix } \mathbb{C}$. The harmonic balance a_k and the geometric eigenvalue $2\cos(2\pi k/12)$ are read off the *same* eigenvector — the question of §1, answered at the abelian floor.

Every abelian Cayley graph. The keystone is not special to C_{12} . For any connection set $S \subseteq \mathbb{Z}_N$, the circulant `cayleyAdj` S of its indicator $\mathbf{1}_S$ is diagonalized by the DFT with eigenvalue **Babai’s character sum** [2] $\hat{\mathbf{1}}_S(k) = \sum_{s \in S} \text{stdAddChar}(-(sk))$ (`cayleyAdj_eigenvalue`), and for a *symmetric* $S = -S$ with $0 \notin S$ it equals Mathlib’s abelian Cayley adjacency circulantGraph S (`cayleyAdj_eq_adjMatrix`). The chromatic cycle is the instance $S = \{1, -1\}$. This is the cycle-graph keystone harvested into the reusable abelian-Cayley-spectrum lemma.

The DFT diagonalizes everything that commutes with transposition. But music also inverts — it reads a chord upside down. Does the eigenbasis survive when we adjoin I , or does the clean diagonal shatter?

4 Adjoin inversion: the dihedral block decomposition

So far only *transposition* T_a has acted. The **inversion** $I : x \mapsto -x$ flips the chromatic circle, and T with I generate the dihedral T/I -group D_{2N} . Directly on the characters,

$$\begin{aligned} I \cdot \text{dftChar } k &= \text{dftChar } (-k) & (\text{inv_dftChar}), \\ T_a \cdot \text{dftChar } k &= \text{stdAddChar}(-(ak)) \cdot \text{dftChar } k & (\text{transl_dftChar}) : \end{aligned}$$

translation is diagonal; inversion permutes the characters by negating the frequency. So D_{2N} only ever pairs k with $-k$. Hence the 2-dimensional block $\text{block } k = \text{span}\{\text{dftChar } k, \text{dftChar } (-k)\}$ is closed under inversion and every transposition (**block_dihedral_invariant**) — the block-diagonalization of the dihedral action in the DFT basis, indexed by the **conjugate pairs** $\{k, -k\}$. For $k \neq -k$ the block is **irreducible** (`block_irreducible`, by eigenvalue separation): the genuine 2-dimensional dihedral irreducibles, realized in the Fourier basis. This is why the “generative $\mathbb{C}[D_{2N}]$ ” program collapses to the abelian pitch-class DFT *quotiented by inversion*.

The tritone, again. The block is 1-dimensional exactly when $k = -k$, i.e. $k \in \{0, N/2\}$: for odd N only the DC term, but for *every even* $N = 2m$ also the **tritone** $N/2$, fixed by inversion in every even universe (`tritone_selfPaired`). The number of 1-dimensional blocks is $\gcd(2, N)$. The tritone is the unique nonzero self-conjugate frequency — the same T_6 that centres the 6-30 self-duality of Note #1 and the index-vector parity of Note #2. All of §4 holds for *every* N .

Any equal temperament, for free. Because every theorem above is stated over an arbitrary $\mathbb{Z}_N$, the block decomposition instantiates in *any* equal temperament with no further proof — a concrete payoff of formalizing over a parameter rather than by hand. In the microtonal systems 19-, 31- and 53-TET (all odd) there is no tritone at all: the only self-paired frequency is the DC term, $\gcd(2, N) = 1$, so a single 1-dimensional block and $(N - 1)/2$ two-dimensional irreducible blocks (`block_irreducible` is generic). In an even temperament such as 24-TET (quarter tones) the tritone returns at $N/2 = 12$, giving two 1-dim blocks. The proof assistant confirms each by one application of the generic census — e.g. $(\{k : \mathbb{ZMod } 19 \mid k = -k\}).\text{nCard} = 1$ and `tritone_selfPaired 12` for 24-TET — all machine- checked in `InversionDFT.lean`, the whole family supplied by the one universe-generic theorem.

The unification with Note #2 (machine-checked). For a real 0/1 indicator the DFT is Hermitian, $\hat{A}(-t) = \overline{\hat{A}(t)}$ (`Ahat_neg`), so the power spectrum is **constant on conjugate pairs**, $|\hat{A}|^2(-t) = |\hat{A}|^2(t)$ (`powerSpec_neg`) — it descends to the inversion quotient $\mathbb{Z}_N/(t \sim -t)$, the very set of conjugate pairs. The interval vector is its fold over those pairs, $\text{IVraw}(A, k) = \mathcal{F}^{-1}(|\hat{A}|^2)(k) + \mathcal{F}^{-1}(|\hat{A}|^2)(-k)$ for $1 \leq k \leq 5$ (`IVraw_eq_powerSpec_conjPair`). So Note #2’s

phase-blind floor (homometry = equal $|\hat{A}|^2$) and Note #3's dihedral blocks are *the same conjugate-pair object*. The two halves are classical — the Hermitian redundancy [1, 15] and the dihedral irreducibles indexed by $\{k, -k\}$ [3]; what we observe is that the two well-known indexings are one object.

The $\pm k$ blocks are abstract representation theory — a bookkeeping of irreducibles. Is there a concrete musical object, a graph one could draw on the Tonnetz, whose adjacency spectrum simply is this block structure made audible?

5 Main theorem: the Tonnetz spectrum is the triad's balance profile

The last rung leaves the abelian world, and lands on Theorem 1. The neo-Riemannian *PLR-group* — generated by the *Parallel*, *Leading-tone* and *Relative* involutions on the 24 major/minor triads — is itself $\cong D_{12}$ (Crans–Fiore–Satyendra [3]). Its Cayley graph Γ is the **Tonnetz**, the chicken-wire torus of Douthett–Steinbach [7]: 24 vertices, 3-regular. The adjacency operator A is symmetric and 3-regular, with two rational eigenvalues: $+3$ (all-ones, the trivial irrep) and -3 (the major/minor *sign* vector — every *PLR* edge crosses major $\leftrightarrow$ minor, so Γ is **bipartite on quality**). The rest is governed by the dihedral irreducibles — and here a *general* fact does the work, of which the Tonnetz is one instance.

Proposition 1 (Dihedral Cayley spectrum; Gao–Luo [10], cf. Liu's survey [11]). *Let $S \subseteq \mathbb{Z}_N$ and form the Cayley graph of D_{2N} on the reflection connection set $\{s r^c : c \in S\}$. Its adjacency spectrum is*

$$\pm \{ |\hat{\mathbf{1}}_S(k)| : k = 0, 1, \dots, \lfloor N/2 \rfloor \}, \quad \hat{\mathbf{1}}_S(k) = \sum_{c \in S} \omega^{kc}, \quad \omega = e^{2\pi i/N},$$

the $\pm$ from bipartiteness (every reflection edge swaps rotations and reflections).

Proof (conceptual). On the 2-dimensional irrep ρ_k ($r \mapsto \text{diag}(\omega^k, \omega^{-k})$, $s \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$) the operator $\sum_{c \in S} \rho_k(s r^c)$ is the off-diagonal block $\begin{pmatrix} 0 & \hat{\mathbf{1}}_S(k) \\ \hat{\mathbf{1}}_S(k) & 0 \end{pmatrix}$, with eigenvalues $\pm |\hat{\mathbf{1}}_S(k)|$; the one-dimensional irreps give $\pm |\hat{\mathbf{1}}_S(0)|$ and $\pm |\hat{\mathbf{1}}_S(N/2)|$. The Cayley spectrum is the union over irreps (Babai [2]). $\square$

This is standard; we record it only to place the music. **Theorem 1 is the case $N = 12$, $S = \{P, L, R\}$.** The *PLR* connection set, in root-shift coordinates, is a transposition of the generating triad $T = \{0, 3, 7\}$, so $\hat{\mathbf{1}}_S(k) = |a_k(T)|$ and the spectrum is the triad's own balance multiset

$$\pm \{ 3, \sqrt{5}, 2 \cos(\frac{\pi}{12}), \sqrt{3}, 1, 2 \cos(\frac{5\pi}{12}) \},$$

characteristic polynomial $(x - 3)(x + 3)(x - 1)^3(x + 1)^3(x^2 - 5)^2(x^2 - 3)^2(x^4 - 4x^2 + 1)^2$, with the **golden** $\sqrt{5} = |1 + 2i| = |a_3|$ the augmented balance. The conceptual reason the Tonnetz sounds like its triad is Proposition 1: *a chord is its graph's connection set*.

Homometry $\Rightarrow$ cospectral (a dictionary entry). By Proposition 1 the spectrum depends on S only through $\{|\hat{\mathbf{1}}_S(k)|\}$ — the interval vector of S . So two pitch-class sets that are **homometric** (Z -related: equal interval vector, equal $|\hat{\mathbf{1}}|$) generate *cospectral* dihedral Cayley graphs. The implication is old — it is Babai's difference-multiset cospectrality criterion ([2], Cor. 4.2), used to manufacture cospectral dihedrants by Abdollahi–Janbaz–Ghahramani [12]; what is new here is only the *identification* of that difference multiset with the music-theoretic interval vector, tying

the spectral fibre to the Z -relation of Note #2 [21]. For instance the all-interval tetrachords 4-Z15 $\{0, 1, 4, 6\}$ and 4-Z29 $\{0, 1, 3, 7\}$ are homometric yet not T/I -related — hence cospectral but non-isomorphic.

Machine-checked. Every eigenvalue is certified in Lean by an explicit eigenvector $Au = \lambda u$ (`A_mulVec_golden`, `A_mulVec_j4/j2/j1/j5`), the negative half free from the quality flip (`bipartite_neg_eigen`); and *completeness* — that these 24 are *all* of them — is machine-checked too (`tonnetz_spectrum_complete_unconditional`): the eigenvectors form an explicit *basis* of $\mathbb{C}^{\text{Triad}}$, so the spectrum is exactly the multiset $\{\pm |\hat{\mathbf{1}}_T(k)|\}_{k=0}^{11}$ — a full diagonalization, built from the “(DFT-character basis) $\otimes$ (quality pair)” independence lemma, with *no* Sage and *no* D_{12} irrep classification (which Mathlib lacks). *Evaluating* that multiset to the six named closed forms with multiplicities 1, 2, 2, 3, 2, 2 — the factored characteristic polynomial of Theorem 1 — is the exact Sage witness (`tonnetz_cayley_spectrum.py`): Lean certifies the basis, Sage the named values.

The musical reading. Each eigenvalue now measures one harmonic periodicity of the generating triad — $\sqrt{5} = |a_3|$ **augmented** (the largest non-trivial balance), $\sqrt{3} = |a_4|$ **diminished-seventh**, $2 \cos(\frac{\pi}{12}) = |a_5|$ **diatonic**, $2 \cos(\frac{5\pi}{12}) = |a_1|$ **chromatic**, $1 = |a_6|$ **whole-tone**, $\pm 3 = |a_0|$ cardinality. (The labels are the standard Quinn–Amiot dictionary, fixed by which scale maximizes each $|a_k|$: augmented triad $\rightarrow a_3$, diminished seventh $\rightarrow a_4$, whole-tone scale $\rightarrow a_6$.) Lostanlen [14] diagonalized the same graph’s *Laplacian* $L = D - A$ (eigenvalues $3 - \lambda$, not the adjacency values) and used its eigenvectors as wavelet filters, never naming the eigenvalues nor connecting them to balances; reading the adjacency spectrum as the triad’s balance profile appears unstated.

So the spectrum is six numbers, $\pm\{3, \sqrt{5}, \sqrt{3}, 1, 2 \cos \frac{\pi}{12}, 2 \cos \frac{5\pi}{12}\}$ — a list. But a list hides a shape. What does the eigenstructure actually look like once the rotations are allowed to turn?

6 The geometry of the spectral structure

The whole ladder has a single geometric face: **the Fourier basis is a system of rotations**, and every structural fact is a feature of how those rotations turn, fit, and refine (companion scripts in `viz/`).

Transposition is a gear train (Figure 1). Transposition multiplies a_k by $e^{-2\pi i k t / 12}$, so coordinate a_k **rotates at speed** k , visiting $12 / \gcd(k, 12)$ positions before it repeats; all gears re-synchronise at the octave $t = 12$. The DC term a_0 has speed 0 — it never moves (the cardinality); the tritone a_6 is real, so it merely flips sign; each conjugate pair is a gear and its mirror, counter-rotating at $\pm k$. The gears mesh into one closed clockwork precisely because the speeds are **integers** (commensurate $\Rightarrow$ periodic); in an irrational tuning the same flow would be ergodic, never re-meshing. The radius of each gear is $|a_k|$ — the triad’s balance, i.e. the Tonnetz eigenvalue of §5.

Inversion folds the basis onto a torus (Figure 2). The Chinese Remainder Theorem $\mathbb{Z}_{12} \cong \mathbb{Z}_3 \times \mathbb{Z}_4$ ($j \mapsto (j \bmod 3, j \bmod 4)$) splits the universe into two factor circles — the **augmented** axis $\langle 4 \rangle = \{0, 4, 8\}$ (the $\mathbb{Z}_3$ factor) and the **diminished** axis $\langle 3 \rangle = \{0, 3, 6, 9\}$ (the $\mathbb{Z}_4$ factor) — and a chromatic semitone is the diagonal step. On the dual side the split is exact, with a small but real index flip worth stating precisely.

Proposition 2 (CRT coordinates of the characters). *Under $\mathbb{Z}_{12} \cong \mathbb{Z}_3 \times \mathbb{Z}_4$, a character χ_k factors through the $\mathbb{Z}_3$ projection iff $k \in \{0, 4, 8\}$, and through the $\mathbb{Z}_4$ projection iff $k \in \{0, 3, 6, 9\}$. Hence χ_4 generates the augmented ($\mathbb{Z}_3$) axis and χ_3 generates the diminished ($\mathbb{Z}_4$) axis, with the tritone χ_6 the order-2 element of the $\mathbb{Z}_4$ part.*

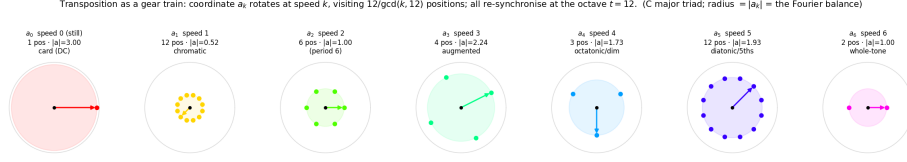


Figure 1: Transposition as a gear train: a_k rotates at speed k , visiting $12/\gcd(k, 12)$ positions; all re-synchronise at the octave. Radius $= |a_k|$, the Fourier balance (a_0 still; a_3 augmented $= \sqrt{5}$; a_4 diminished $= \sqrt{3}$; a_6 whole-tone; the rest as labelled).

Proof. χ_k factors through $\pi_3 : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_3$ iff it is trivial on $\ker \pi_3 = \{0, 3, 6, 9\}$, i.e. $\chi_k(3) = e^{\pi i k/2} = 1$, i.e. $4 \mid k$; symmetrically through π_4 iff $\chi_k(4) = e^{2\pi i k/3} = 1$, i.e. $3 \mid k$. $\square$

The flip is the point: the augmented triad *maximises* $|a_3|$ because χ_3 is *constant* on it (it factors through the $\mathbb{Z}_4$ quotient), whereas the character that *resolves* the augmented axis is χ_4 . So “which scale saturates a_k ” and “which χ_k coordinatises the factor” read off with the indices $3 \leftrightarrow 4$ swapped — two dual faces of one CRT splitting. *Within* a factor circle the points are locked; the diagonal *couples* the two factors into the single chromatic clock — the same factor-vs-diagonal that the conjugate-pair blocks of §4 express as counter-rotation vs common drive.

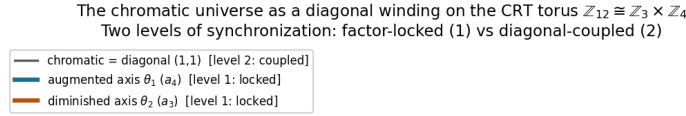


Figure 2: $\mathbb{Z}_{12} \cong \mathbb{Z}_3 \times \mathbb{Z}_4$ as a diagonal winding on the CRT torus: the factor circles are the augmented ($\mathbb{Z}_3$, resolved by χ_4) and diminished ($\mathbb{Z}_4$, resolved by χ_3) axes (Prop. 2); the chromatic is the diagonal. Two levels of synchronisation: factor-locked and diagonal-coupled.

Why the tritone seeds every halving. Before the geometry, the algebra that drives it.

Model the pitch-class circle as $S^1 = \mathbb{R}/\mathbb{Z}$; the 2^n -tone equal temperament is the subgroup $\mu_{2^n} \cong \mathbb{Z}_{2^n}$ of 2^n -th roots of unity, and octave bisection nests them $\mathbb{Z}_2 \subset \mathbb{Z}_4 \subset \mathbb{Z}_8 \subset \dots$. The map linking one level to the next is the **doubling** $\delta : z \mapsto z^2$ (on angles $\theta \mapsto 2\theta$), the 2-to-1 octave-folding of the circle. It is exactly 2-to-1: the fibre of δ over the unison $z = 1$ is $\{z : z^2 = 1\} = \{+1, -1\}$, and $-1 = e^{\pi i}$ is the **tritone**, the unique order-2 element of S^1 (**tritone_selfPaired**, at every even level). So the binary question at each stage — “*which square root of the unison?*” — is precisely “*unison or tritone?*”: the tritone is the nontrivial element of every bonding fibre.

The dyadic tower: one theorem, and an honest analogy (Figure 3). Iterating the fibre, the kernels $\ker \delta^n = \mu_{2^n} \cong \mathbb{Z}_{2^n}$ form a genuine projective system under doubling, and its inverse limit is a *theorem*, not a metaphor: $\varprojlim (\dots \rightarrow \mathbb{Z}_{2^{n+1}} \rightarrow \mathbb{Z}_{2^n}) \cong \mathbb{Z}_2$, the 2-adic integers — the Cantor fibre of the Smale–Williams solenoid $\mathbb{S}_2 = \varprojlim (S^1 \xleftarrow{\delta} S^1 \dots)$, with the tritone the order-2 element at each finite stage and none surviving the torsion-free limit (geometrically, δ wraps the solid-torus tube twice around its core; the nested intersection $\bigcap_n \delta^n(\text{tube})$ is the attractor). **What we do not claim** is a theorem identifying the §4 block decomposition — which lives at a single fixed N — *with* this solenoid. The tempting route fails cleanly, and we record the negative: the self-paired characters $\chi_{2^{n-1}}$ are *all* the one sign character $(-1)^j$, so the naive tower of self-paired subspaces $V_n = \langle \chi_0, \chi_{2^{n-1}} \rangle$ is constant and $\varprojlim V_n$ is merely $\mathbb{C}^2$, not $\mathbb{S}_2$. We therefore keep the solenoid as a *picture* of the dyadic-grid tower, not as a corollary of the block theorem; the precise bridge — a projective system built from the block data and converging to $\mathbb{S}_2$ — we flag as **open** (§8). *The solenoid and 2-adic-in-music are classical (Pitkänen [16]; Fokker [8]; the $\mathbb{Z}_4$ /diminished torus is in Yust [19]); we add only the doubling-fibre reading of the tritone and the explicit negative above.*

The Smale–Williams solenoid: a torus inside a torus, wrapping twice, iterated
 core = the diminished $\mathbb{Z}_4$ cycle; doubling = halving the octave ($\mathbb{Z}_4 \subset \mathbb{Z}_8 \subset \dots$) = the 2-adic tower; the limit is fractal (Cantor $\times$ arc)

— C_0 : core = dim7 cycle $\mathbb{Z}_4$ (1 wrap, 4-EDO)
 — C_1 : 2 wraps (8-EDO)
 — C_2 : 4 wraps (16-EDO)
 — C_3 : 8 wraps (32-EDO)

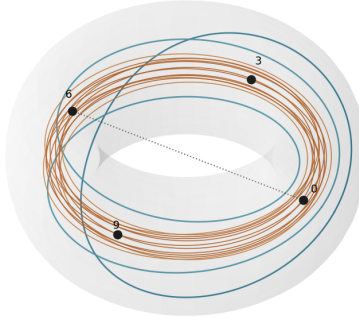


Figure 3: The Smale–Williams solenoid: a torus wound twice inside a torus, iterated. The doubling $\delta : z \mapsto z^2$ is 2-to-1, its fibre over the unison being {unison, tritone}; the inverse limit $\varprojlim (S^1 \xleftarrow{\delta} S^1 \dots)$ has a Cantor ($\mathbb{Z}_2$) fibre built from those binary choices. We use it as a picture of the dyadic-grid tower, not as a corollary of the block theorem (see text).

Every step so far has claimed a theorem. But a claim is not a proof, and a diagram is not a

certificate. Which of these does a proof kernel actually verify — and at what axiomatic cost?

7 What is machine-checked

The spectral ladder — now including completeness — is certified by the Lean 4 kernel across five sorry-free files, each with an axiom audit (`#print axioms`) returning at most `[propext, Classical.choice, Quot.sound]` and no `sorryAx`; several decidable witnesses are choice-free. Verified by recompiling each from scratch in the godsil fast-loop environment (EXIT 0).

Theorem	File	Statement
<code>circulant_mulVec_dftChar</code>	<code>CycleGraphSpectrum</code>	Every circulant diagonalized by the DFT; eigenvalue = $\hat{v}(k)$ (keystone).
<code>C12_eigenvalue_cos</code>	<code>CycleGraphSpectrum</code>	The C_{12} spectrum is $2 \cos(2\pi k/12)$.
<code>C12_eq_adjMatrix</code>	<code>CycleGraphSpectrum</code>	C_{12} is Mathlib’s <code>cycleGraph 12</code> adjacency matrix.
<code>cayleyAdj_eigenvalue</code>	<code>CycleGraphSpectrum</code>	Abelian Cayley spectrum = $\sum_{s \in S} \text{stdAddChar}(-(sk))$ (Babai).
<code>inv_dftChar / transl_dftChar</code>	<code>InversionDFT</code>	Inversion swaps $k \leftrightarrow -k$; translation acts diagonally.
<code>block_dihedral_invariant</code>	<code>InversionDFT</code>	The D_{2N} action is block-diagonal (conjugate-pair blocks).
<code>block_irreducible</code>	<code>InversionDFT</code>	Each 2-dim block ($k \neq -k$) is an irreducible dihedral irrep.
<code>tritone_selfPaired</code>	<code>InversionDFT</code>	The tritone is inversion-fixed in every even universe.
<code>powerSpec_neg</code> / <code>IVraw_eq_powerSpec_conjPair</code>	<code>Fourier</code>	$ \hat{A} ^2$ constant on conjugate pairs; the interval vector is its fold over $\{k, -k\}$ (the Note #2 weld).
<code>A_mulVec_ones / A_mulVec_sign</code>	<code>TonnetzSpectrum</code>	Tonnetz eigenvalues $+3$ and -3 (bipartite quality flip).
<code>A_mulVec_golden, _j4/j2/j1/j5</code>	<code>TonnetzSpectrum</code>	The full irrational spectrum $\sqrt{5}, \sqrt{3}, 1, 2 \cos(\frac{\pi}{12}), 2 \cos(\frac{5\pi}{12})$, by explicit eigenvectors.
<code>tonnetz_spectrum_complete_unconditional</code>	<code>TonnetzCompleteness</code>	The 24 eigenvectors form a <i>basis</i> — completeness: the spectrum is exactly $\{\pm \hat{1}_T(k) \}$ (no Sage, no irreps). The named values + multiplicities are the Sage witness.

Table 1: The spectral ladder, machine-checked across `CycleGraphSpectrum.lean`, `InversionDFT.lean`, `Fourier.lean`, `TonnetzSpectrum.lean` and `TonnetzCompleteness.lean`. Mathlib carries `circulant`, `ZMod.dft`, `cycleGraph/circulantGraph` and the dihedral group as *definitions* but no spectrum theorem and no dihedral-irrep classification; these files supply the spectral layer.

Closing the last formal gap: a Mathlib roadmap

The spectrum is *complete* in Lean without the irrep classification (Table 1, last row): a DFT-character basis tensored with a quality two-vector suffices. What remains open is only the *named* isotypic statement — “ $\mathbb{C}[D_{2N}] \cong \bigoplus_k \rho_k$ as D_{2N} -modules” — which Mathlib cannot yet phrase because it has `DihedralGroup N` and `Representation` as definitions but no classification of dihedral irreducibles. It is a coverage gap, not a hard theorem; the route is three scoped bricks, in dependency order:

1. **The dihedral irreducibles, named.** Build $\rho_k : \text{DihedralGroup } N \rightarrow \text{Matrix } (\text{Fin } 2) \mathbb{C}$ ($r \mapsto \text{diag}(\omega^k, \omega^{-k})$, $s \mapsto \text{antidiagonal}$) and prove `IsSimpleModule` for $k \neq -k$. The hard half is already done — `block_irreducible` is exactly “no proper invariant subspace”; the brick is the explicit intertwiner block $k \simeq_{D_{2N}} \rho_k$, the constant matrix $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$.
2. **Schur over $\mathbb{C} \Rightarrow$ isotypic `DirectSum`.** With ρ_k named and `block_dihedral_invariant` giving the block decomposition, Mathlib’s finite-dimensional Schur (`Module.End` of a simple module is a division ring) packages the regular representation as the internal `DirectSum` $\bigoplus_k$ block k with multiplicities $\gcd(2, N)$ ones and the rest twos (our `selfPaired_ncard_even`).
3. **$D_{2N}/Z \cong D_N$ for the centralizer (Note #1 tie).** The quotient by the centre closes the named `Cgrp` $\cong D_6$ of the 6-30 brick; Mathlib currently offers only `center_eq_bot_of_odd_ne_one`, so the even case is the missing lemma.

Each is standard representation theory; none needs new mathematics. We flag them as the natural upstreaming targets — the point at which this note’s spectral layer would become reusable Mathlib infrastructure rather than project-local files.

8 Novelty, scope, and what this note does not claim

Contribution, stated plainly. This is a formalization-and-organization note, not new mathematics. Circulant and Cayley-graph spectra are folklore (Godsil–Royle [9]; Babai [2]); the $\pm k$ pairing of dihedral irreducibles is standard representation theory [6]; the closed-form dihedral Cayley spectrum is Gao–Luo [10] (of which the Tonnetz, Theorem 1, is the $S = \{P, L, R\}$ instance); the Tonnetz Laplacian was diagonalized by Lostanlen [14], the $PLR \cong D_{12}$ duality is Crans–Fiore–Satyendra [3], the graph is Douthett–Steinbach [7], the balance dictionary is Quinn–Amiot [18, 1]. What is **new** is the *unified, machine-checked* account, and within it the **first formalization in any proof assistant** of: (i) the circulant diagonalization-by-DFT and the abelian-Cayley spectrum; (ii) the DFT-basis block decomposition of the D_{2N} action and the 2-dimensional block irreducibility — the dihedral representation theory on pitch classes; (iii) explicit eigenvectors certifying the entire Tonnetz spectrum — the $S = \{P, L, R\}$ instance of the Gao–Luo formula [10], here machine-checked. All are made universe-generic. Beyond formalization, one *expository* identification appears unstated: (iv) that the Tonnetz adjacency spectrum is the triad’s own Fourier-balance multiset (§5); its components are classical, the weld is the contribution.

What this note does *not* claim.

- **Not new mathematics.** First *formalization* and *organization* only.
- **Not a new formalization of the T/I action.** Prismriver [17] (Lean 4) formalizes the bare D_{2N} action on pitch classes; we cite it and claim only the Fourier-basis *representation theory* and the circulant/Cayley spectrum, which it does not treat (confirmed by a file-by-file audit: no DFT/character/circulant/irrep content).
- **Complete in Lean, but not via the irrep classification.** The spectrum is now *complete* in Lean — the 24 eigenvectors form a certified basis (`tonnetz_spectrum_complete_unconditional`), a full diagonalization — yet *not* by the D_{12} isotypic decomposition (Mathlib lacks it); completeness comes instead from DFT-character $\otimes$ quality-pair independence. The named dihedral-irrep isomorphism remains the one open formal item.

- **Not a golden-ratio mystery — but $\sqrt{5}$ is the augmented balance.** As a bare identity $|1 + 2i| = \sqrt{5}$ occurs whenever $4 \mid n$; its musical content is that $\sqrt{5} = |a_3|$ is the triad’s augmented balance (its largest non-trivial Fourier component), the slowest-mixing harmonic mode (spectral gap $3 - \sqrt{5}$).
- **Not first-contact for 2-adic music.** The dyadic-tower/solenoid remark (§6) is expository; Pitkänen [16] and Fokker [8] precede it, and we cite them.

Open frontier. The named dihedral-irrep isomorphism and the isotypic decomposition (pending a Mathlib classification) — now the *only* remaining formal gap, since spectral completeness is machine-checked (§5); and the literal cross-file welding of the conjugate pair to the index-vector fold of Note #2.

Coda: questions the eigenbasis leaves open

We began with one basis and followed it across three symmetries. The picture it leaves is unreasonably tidy — tidy enough to make one ask what that tidiness is really saying.

If a single basis diagonalizes transposition, inversion, and the neo-Riemannian group at once, is a style’s “colour” just a choice of which Fourier balance to maximize — and could a working theory of harmony be rebuilt, from the balance profile up, rather than from the chord down?

The tritone is the unique self-paired frequency in every even universe — the one fixed point the symmetry cannot move, the singularity that forces the blocks. Is its long history as the diabolus in musica a cultural echo of a structural fact, the ear hearing an algebraic invariant centuries before it could be named?

Completeness needed no D_{12} irrep classification — a character basis tensored with a two-vector sufficed. Where else does a “hard” representation-theoretic theorem quietly dissolve into linear algebra, the moment the right basis is written down?

Data and code availability

The Lean sources `CycleGraphSpectrum.lean`, `InversionDFT.lean`, `TonnetzSpectrum.lean`, `TonnetzCompleteness.lean` (the completeness basis), `Fourier.lean`, the Sage script `tonnetz_cayley_spectrum.py` (load-bearing for the named factored characteristic polynomial and its multiplicities, which Lean does not evaluate — the Lean basis certifies completeness as the multiset $\{\pm|\hat{1}_T(k)|\}$, the Sage witness identifies it with the named values), the microtonal `z24_saturation.sage` (exact cyclotomic), and the figure scripts in `viz/` accompany this note; each Lean theorem is checkable by `#print axioms` and compiles axiom-clean (only `propext`, `Classical.choice`, `Quot.sound`).

A The spectrum as a generative engine

The note reads scales and chords *forward*: a pc-set $A \subseteq \mathbb{Z}_N$ has a Fourier profile $|a_k(A)|$, $a_k(A) = \sum_{j \in A} e^{-2\pi i k j / N}$, and the largest $|a_k|$ names its colour (augmented at $k=3$, octatonic at $k=4$, whole-tone at $k=6$). Composition is the *inverse* problem: fix a colour k , and let the algebra hand back the set that saturates it. This is the practical face of “composing in Fourier space” (Amiot [1], ch. on composition); we record the exact rule and the audio it produced.

The saturation rule (exact). For a set of cardinality d , $|a_k(A)| \leq d$, with equality iff all phases coincide, i.e. $k \cdot A$ is constant in $\mathbb{Z}_N$ — iff A lies in a single coset of

$$H_k = \ker(j \mapsto kj) = \frac{N}{\gcd(N,k)} \mathbb{Z}_N, \quad |H_k| = \gcd(N, k).$$

So the *maximal*- $|a_k|$ set of cardinality $\gcd(N, k)$ is exactly a coset $c + H_k$, and its balance is $|a_k| = \gcd(N, k)$. In $\mathbb{Z}_{12}$ this is precisely Messiaen’s catalogue of *modes of limited transposition* [5]: $H_3 = \{0, 4, 8\}$ the augmented triad, $H_4 = \{0, 3, 6, 9\}$ the diminished seventh, $H_6 = \{0, 2, 4, 6, 8, 10\}$ the whole-tone scale. When $\gcd(N, k) = 1$ (e.g. $k=5, 1$ in $\mathbb{Z}_{12}$), Huddling gives the maximiser at cardinality d as a translate of $k^{-1}\{0, 1, \dots, d-1\}$. For $k=1$ this is a chromatic cluster; for $N=12$, $k=5$, $d=7$ it is the diatonic. An arbitrary unit frequency and cardinality need not select a maximally even set.

Algorithm (reproducible).

1. Pick a colour $k \in \mathbb{Z}_N$ and a length L .
2. If $\gcd(N, k) > 1$: seed $\leftarrow$ a coset $c + H_k$ (a Messiaen mode, $|a_k|$ saturated); else seed $\leftarrow$ a translate of $k^{-1}\{0, \dots, d-1\}$ at the chosen cardinality d .
3. Voice-lead by walking the Tonnetz with conjugate-pair-preserving moves ($\text{PLR} \cong D_{12}$): each triad’s neighbourhood spectrum is *its own* colour profile (§5), so the walk keeps the harmony inside the chosen Fourier band.
4. Emit L pitch-classes traversing the seed and its Tonnetz orbit.

Colour k	$ a_k $	Reference collection	Realisation
3	$\sqrt{5}$ (triad), 3	augmented triad $\{0, 4, 8\}$	augmented_cycle.wav
4	$\sqrt{3}$ (triad), 4	diminished seventh $\{0, 3, 6, 9\}$	petrushka_octatonic.wav
6	1 (triad), 6	whole-tone $\{0, 2, 4, 6, 8, 10\}$	modes_messiaen.wav
5	$2 + \sqrt{3}$	seven-note diatonic (no coset)	tonnetz_hexatonic.wav
even k	2	tritone axis $\{0, 6\}$, the 2-torsion	tritone_symmetry.wav

For frequency five, $2 \cos(\pi/12)$ is the generating triad’s magnitude in Section 5; the seven-note diatonic maximum is $2 + \sqrt{3}$. The table’s sound files are illustrative realizations, not equality certificates for their entire recordings.

Microtonal instance ($N = 24$, quarter tones). The saturation rule is N -generic — H_k and $|a_k| = \gcd(N, k)$ are stated over an arbitrary universe, the same parametrisation the Lean `selfPaired_even/tritone_selfPaired` carry over all N (§4). Reading it at $N = 24$ produces the quarter-tone catalogue of limited-transposition modes, and that catalogue *strictly extends* the chromatic one: the tritone fixed point migrates to $q = N/2 = 12$, and one mode — H_8 — lands on *odd* quarter tones, with no 12-TET analogue.

Audio: `microtonal_z24.wav` sounds the five cosets in 24-EDO (additive sines at $f_0 2^{q/24}$; a quarter tone is not a 12-TET MIDI semitone, so this one is WAV-only). H_8 is the audible payoff — a symmetric mode the chromatic scale cannot voice.

k	$ a_k $	Saturating coset $H_k \subseteq \mathbb{Z}_{24}$ (quarter tones)	12-TET reading
3	3	$\{0, 8, 16\}$	augmented triad (even grid)
4	4	$\{0, 6, 12, 18\}$	diminished seventh (even grid)
6	6	$\{0, 4, 8, 12, 16, 20\}$	whole-tone scale (even grid)
8	8	$\{0, 3, 6, 9, 12, 15, 18, 21\}$	off-grid — a genuinely 24-TET mode
12	12	$\{0, 2, 4, \dots, 22\}$	the full chromatic scale

Scope (honest). The construction is not new mathematics: the coset-saturated sets *are* Messiaen’s limited-transposition modes, and maximal evenness is Clough–Douthett / Amiot. What this appendix adds is the unifying reading — *composition = choosing which Fourier balance to saturate, after which the subgroup H_k (or maximal evenness) dictates the material* — together with the Tonnetz-spectrum tie (§5: a generated triad’s adjacency spectrum returns its own balances) and the reproducible audio. It is an illustration that the forward spectral theory of this note is also generative; it is not a compositional theory in its own right.

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